

Grade 6 Accelerated
Unit 15
Lesson 4 Homework

Name Answer Key
Date _____
Period _____

1. The probabilities of four events are shown. List them in order from least to greatest.

$$\frac{4}{11}$$

5 out of 8

55%

0.46

$$\frac{4}{11}$$

0.46

55%

5 out of 8

Use the spinner to determine missing events, probabilities, and descriptions of likelihood. Express all probabilities as percentages. Use the vocabulary to complete each sentence.

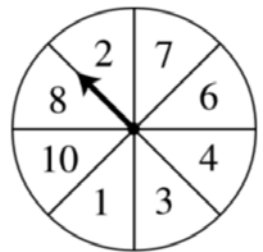
impossible

unlikely

equally likely as
unlikely

likely

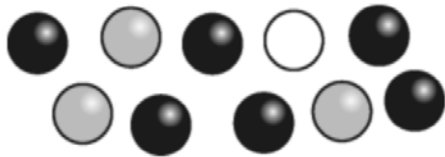
certain



2. The probability of landing on an even number, $P(\text{even})$, is 62.5%. This event is likely.
3. The probability of landing on a number less than 20, $P(\text{less than } 20)$, is 100%. This event is certain.
4. The probability of landing on prime number, $P(\text{prime})$, is 37.5%. This event is unlikely.
5. The probability of landing on a 5, $P(5)$, is 0%. This event is impossible.
6. The probability of landing on a number that is not a 1, $P(\text{not a } 1)$, is 87.5%. This event is likely.

For questions 7-10, determine which event is more likely.

7. Martin put 10 marbles in a jar. The probability he chooses a black marble is 0.6. The probability of choosing a white marble is 10%. Which is more likely to occur?



Black marble: 0.6, white marble: 0.10
Choosing a black marble is more likely.

8. Two students roll the die. Player *A* wins if they roll an even number. Player *B* wins if they roll a number less than 3. Who has the higher probability of winning? Explain.



Player *A*: $\frac{3}{6} = 0.50$

Player *B*: $\frac{2}{6} = 0.\bar{3}$

Player *A* has a greater probability, 0.50, than Player *B*, $0.\bar{3}$.

9. Fort Lauderdale airport reported that on a given day, the probability that an airline arrived on time was 0.435. The likelihood that the next airplane arrived earlier than scheduled was $\frac{1}{6}$ and the later than scheduled was 27%. What is most likely to occur when the next airplane arrives?

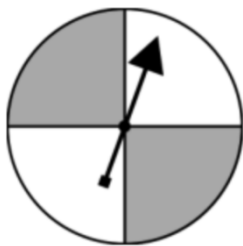
On time: 0.435

Early: $0.1\bar{6}$

Late: 0.270

The airline arriving on time is most likely to occur next.

10. The following spinner is spun. On the next spin, which color is the spinner more likely to land on? Explain.



They are equally likely because the probability is the same for both colors, $\frac{1}{2}$ or 50%.